

Software Engineering

CE2006/CZ2006

UML Review 1

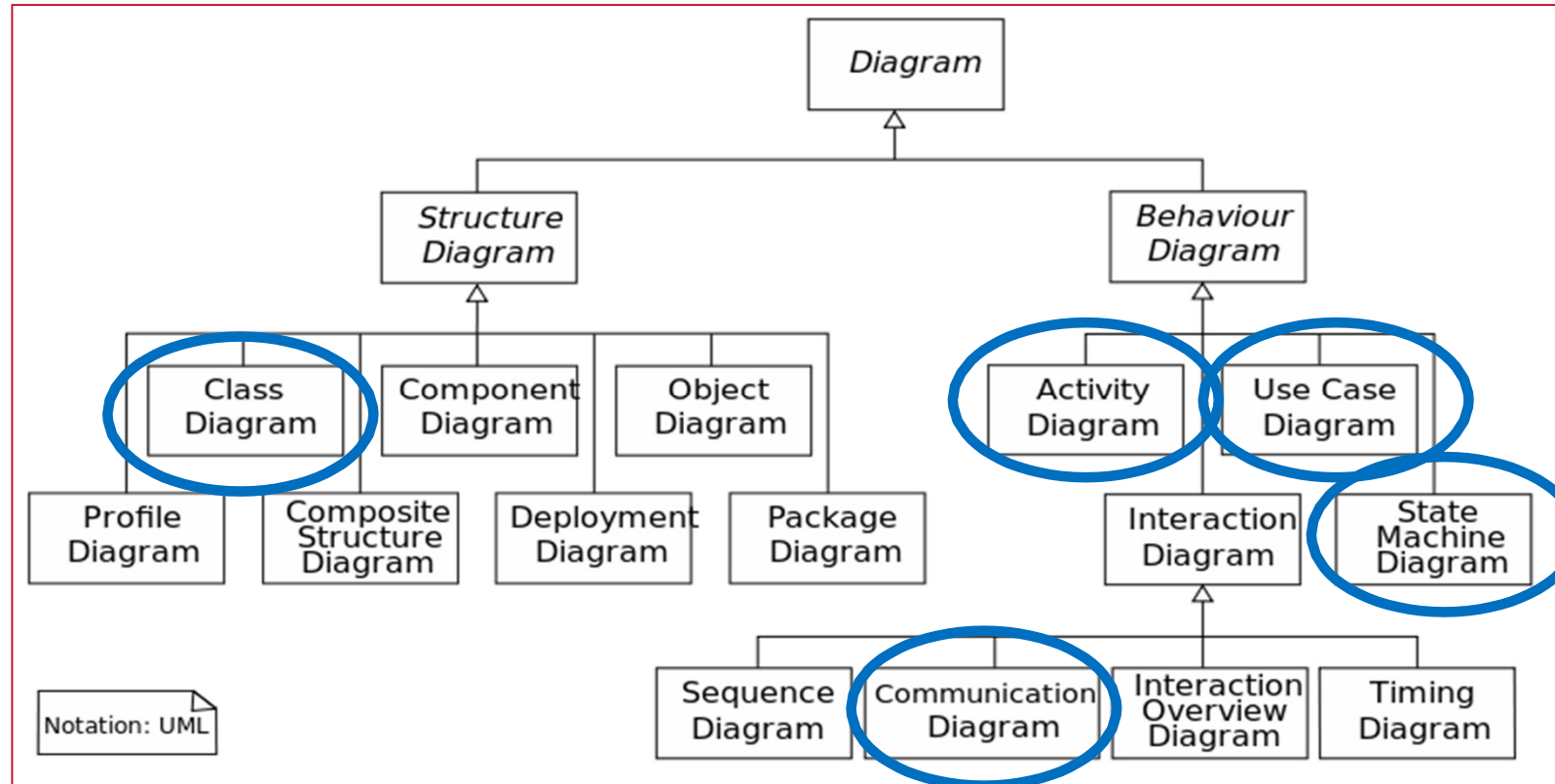
Lesson Objectives

Review UML with an example

At the end of the lesson, you should be able to:

- Describe UML diagrams learned so far
- Explain the purposes of different kinds of UML diagrams
- Use different diagrams at different steps in software requirement elicitation and analysis process

UML diagrams learned



Activity diagram: Flow from one **activity** to another describing the operation of a system.

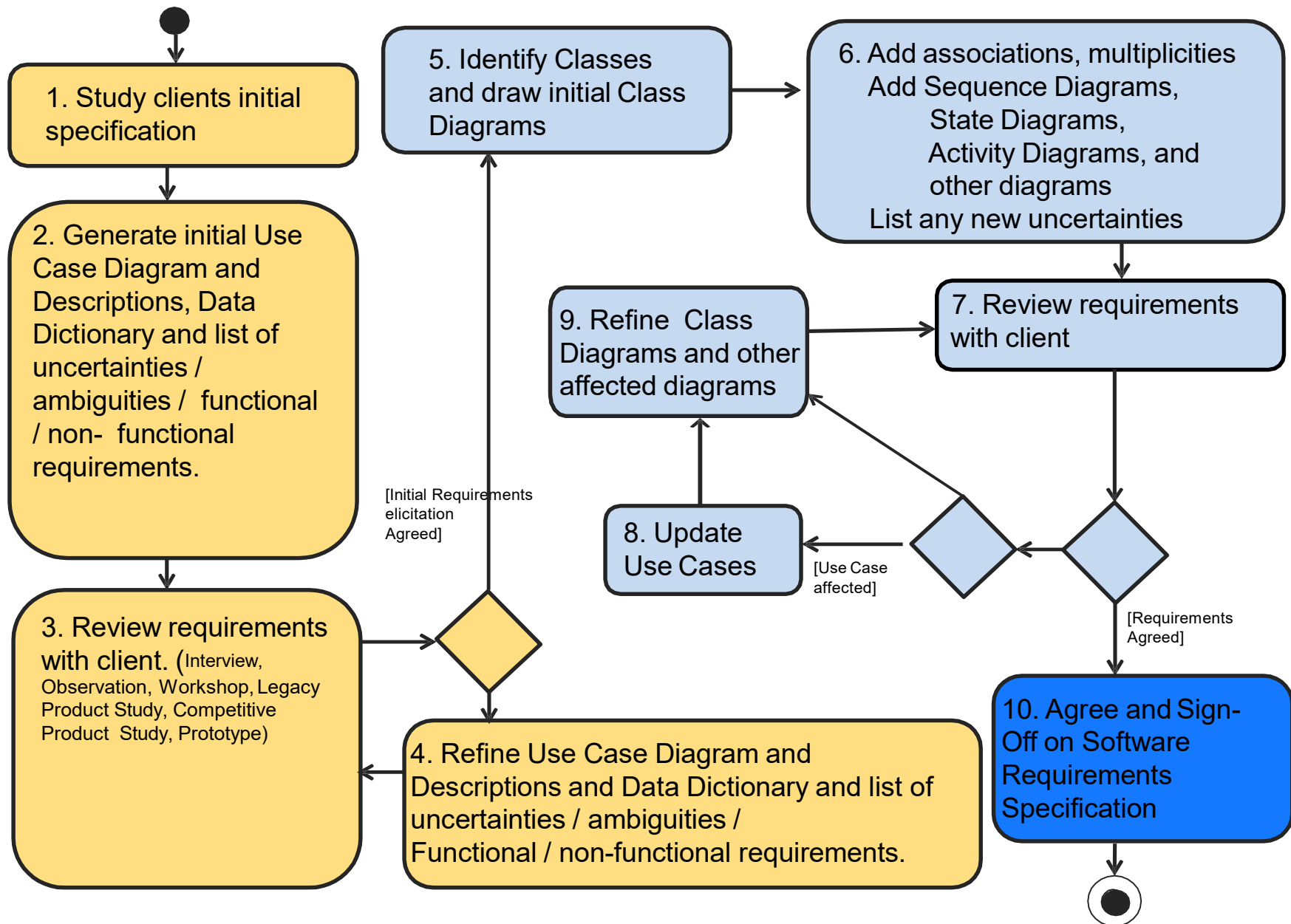
Use Case diagram: describes the **functionality** provided by a system in terms of **actors**, their goals represented as **use cases**, and any dependencies among those use case descriptions.

Class diagram (Static structure diagram): describes the **structure of a system** by showing the system's **classes**, their attributes, and the relationships among the classes.

Sequence diagram and Communication diagram: shows how objects **communicate** with each other in terms of a **sequence** of **messages**.

State machine diagram: describes the **states** and state **transitions** of the system.

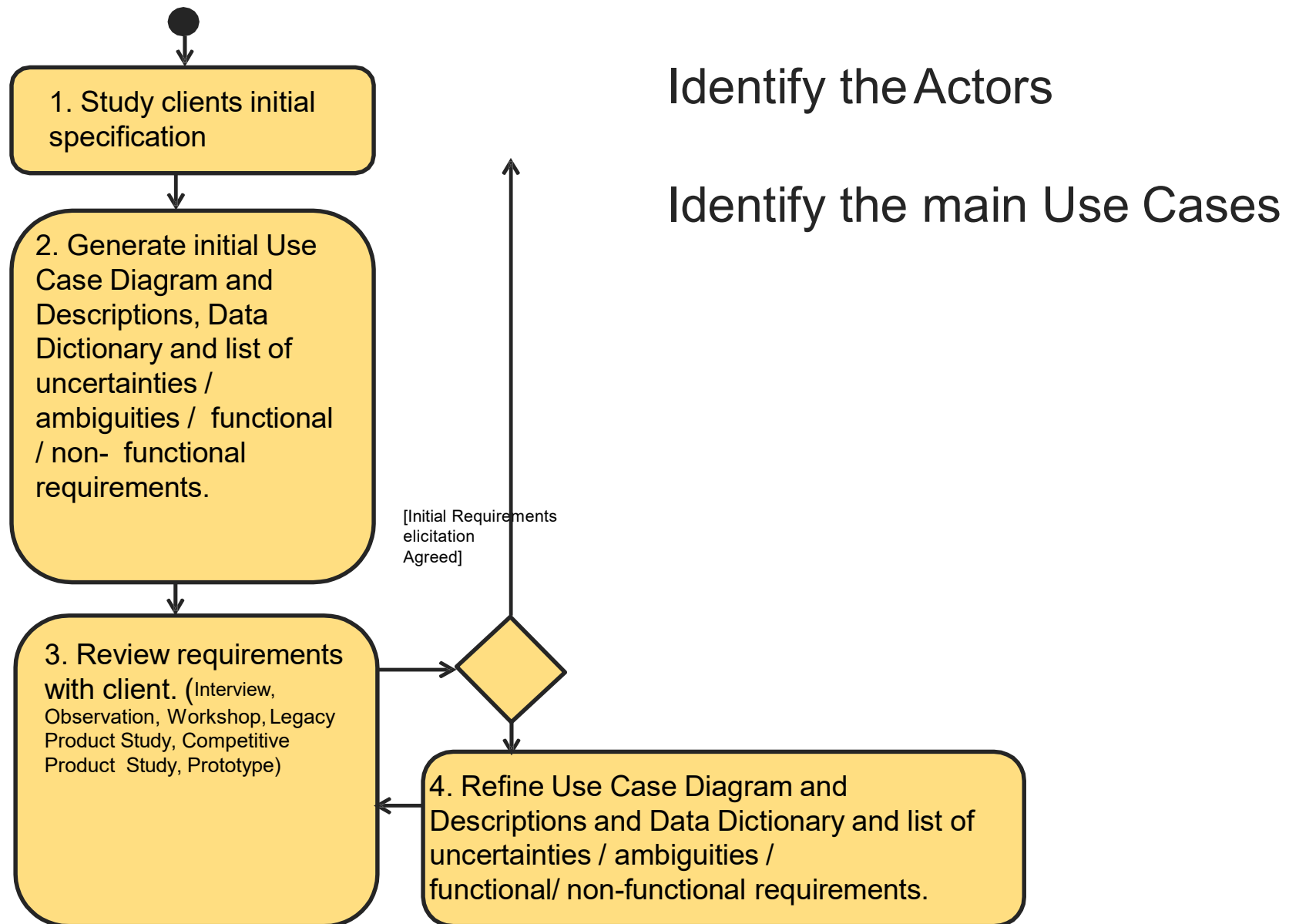
Requirement Elicitation and Analysis Process



Example: A security door access system

1. [A door access system](#) is needed to allow employees to enter a secure area using their fingerprint.
2. When an employee needs to open the door he/she must place their index finger on a fingerprint reader located beside the door.
3. The scanned fingerprint is compared and validated against a company database of employee fingerprints.
4. If a fingerprint match is found the message “Please enter” is displayed and the door is opened to allow the employee to enter.
5. If the fingerprint match fails the message “Access Denied” is displayed and the access system reverts to waiting for a new fingerprint to scan.
6. Every time a door access is attempted a record log is maintained. The information recorded for a successful access is the Time of Access and the name of employee allowed access. The information recorded for a failed access is the Time of Unsuccessful Attempted Access.
7. The company’s Human Resources department maintains a database of employee fingerprints. These are collected once during the new employee enrolment process and are removed when an employee leaves the company.
8. The Human Resources department can view the record log of accesses and attempted accesses and update the employees fingerprint data if necessary.

First Step – Generate Use Case Diagram



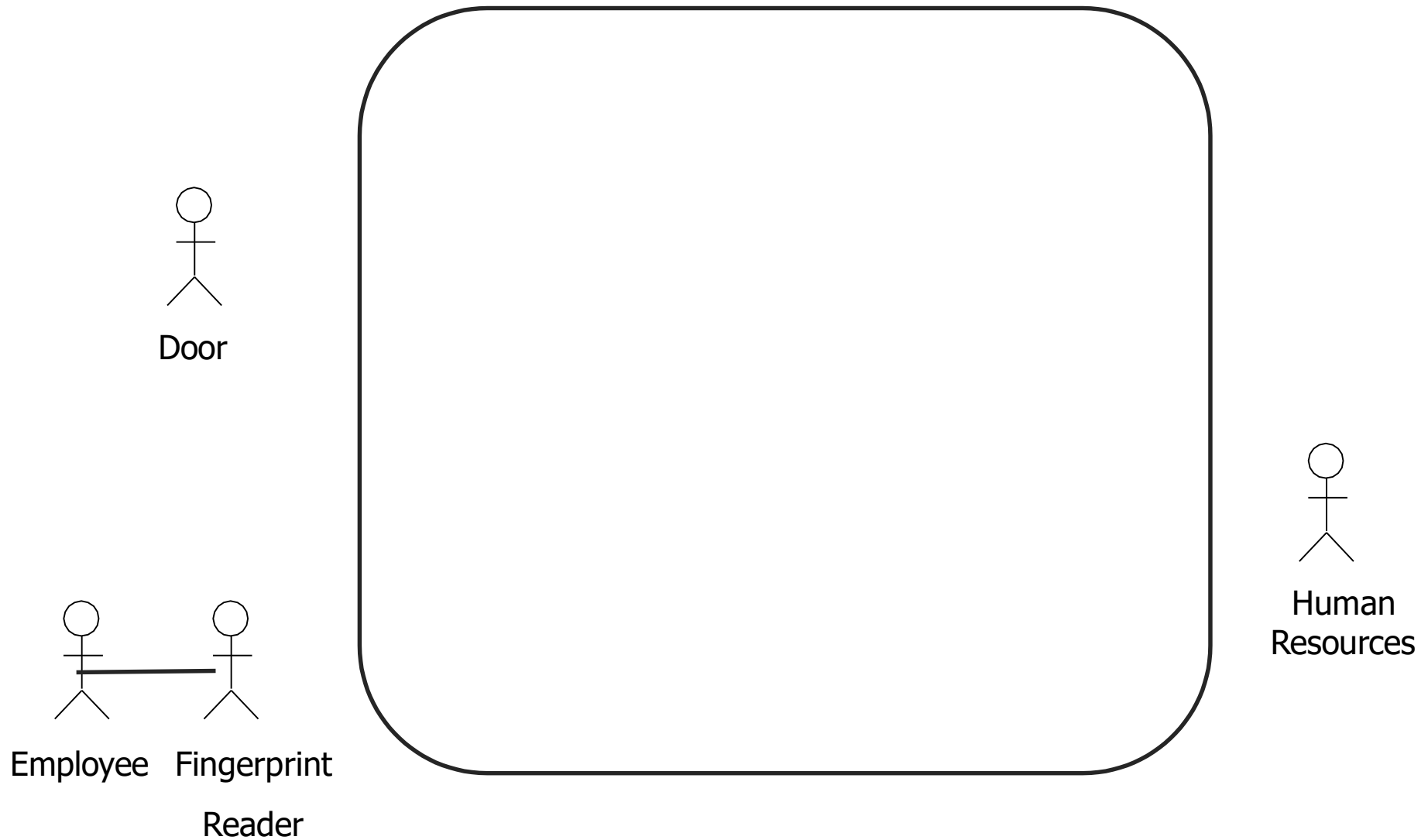
1. Study clients initial specification

IDENTIFY THE ACTORS

1. A door access system is needed to allow employees to enter a secure area using their fingerprint.
2. When an employee needs to open the door he/she must place their index finger on a fingerprint reader located beside the door.
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2. Generate initial Use Case diagram

IDENTIFY THE ACTORS



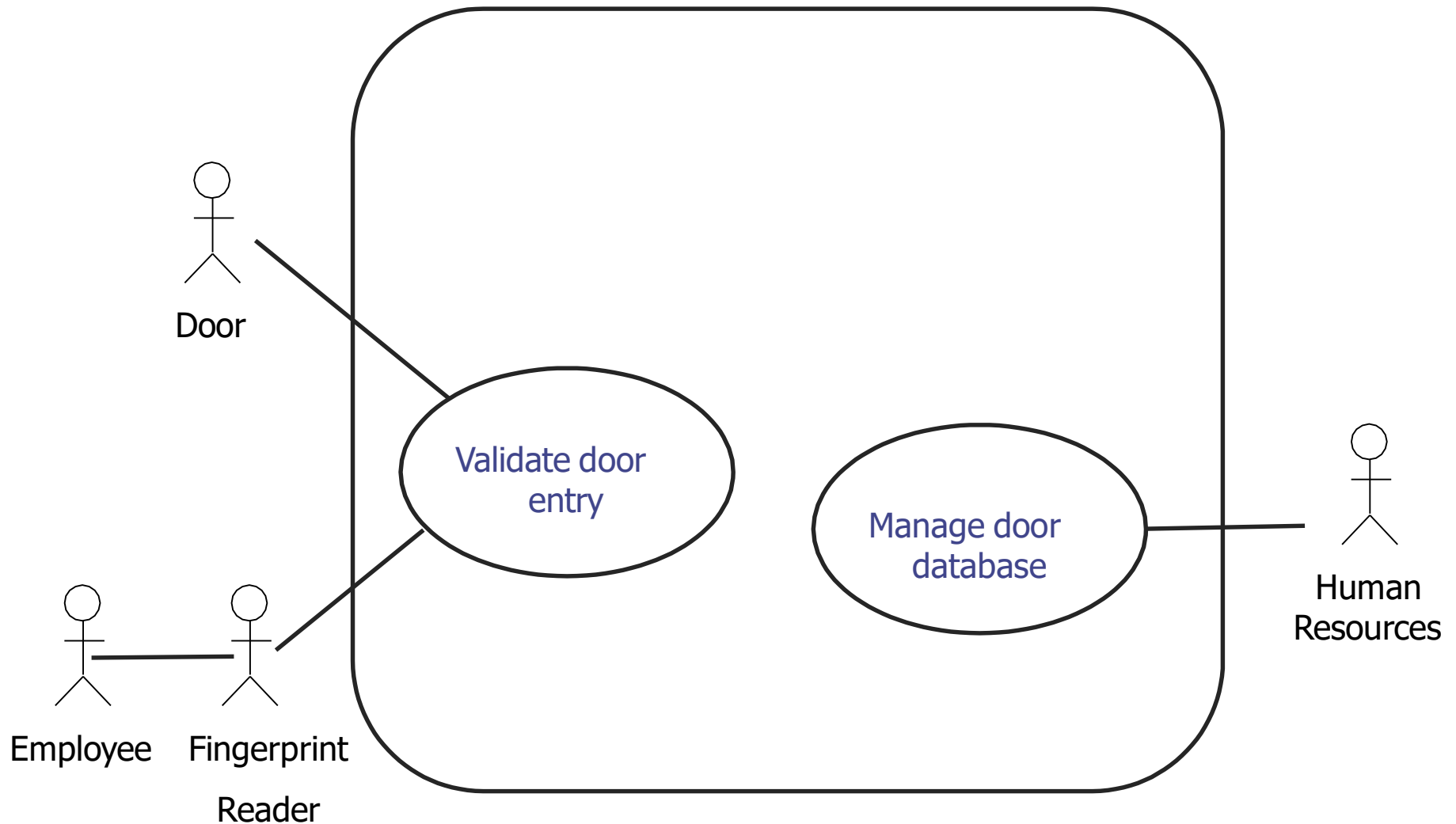
1. Study clients initial specification

IDENTIFY THE MAIN USE CASES (action verb)

1. A door access system is needed to allow employees to enter a secure area using their fingerprint.
2. When an employee needs to open the door he/she must place their index finger on a fingerprint reader located beside the door.
3. The scanned fingerprint is compared and validated against a company database of employee fingerprints.  **Validate**
 **Scan fingerprint**
4. If a fingerprint match is found the message “Please enter” is displayed and the door is opened to allow the employee to enter.
5. If the fingerprint match fails the message “Access Denied” is displayed and the access system reverts to waiting for a new fingerprint to scan.
6. Every time a door access is attempted a record log is maintained. The information recorded for a successful access is the Time of Access and the name of employee allowed access. The information recorded for a failed access is the Time of Unsuccessful Attempted Access.
7. The company’s Human Resources department maintains a database of employee fingerprints. These are collected once during the new employee enrolment process and are removed when an employee leaves the company.  **Add employee**
8. The Human Resources department can view the record log of accesses and attempted accesses and update the employees fingerprint data if necessary.

2. Generate initial Use Case diagram

IDENTIFY THE MAIN USE CASES



2. Generate initial Use Case descriptions

See the Use Case Description Template available on NTULearn course website.

Use Case ID:			
Use Case Name:			
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	
Actor:			
Description:			
Preconditions:			
Postconditions:			
Priority:			
Frequency of Use:			
Flow of Events:			
Alternative Flows:			
Exceptions:			
Includes:			
Special Requirements:			
Assumptions:			
Notes and Issues:			

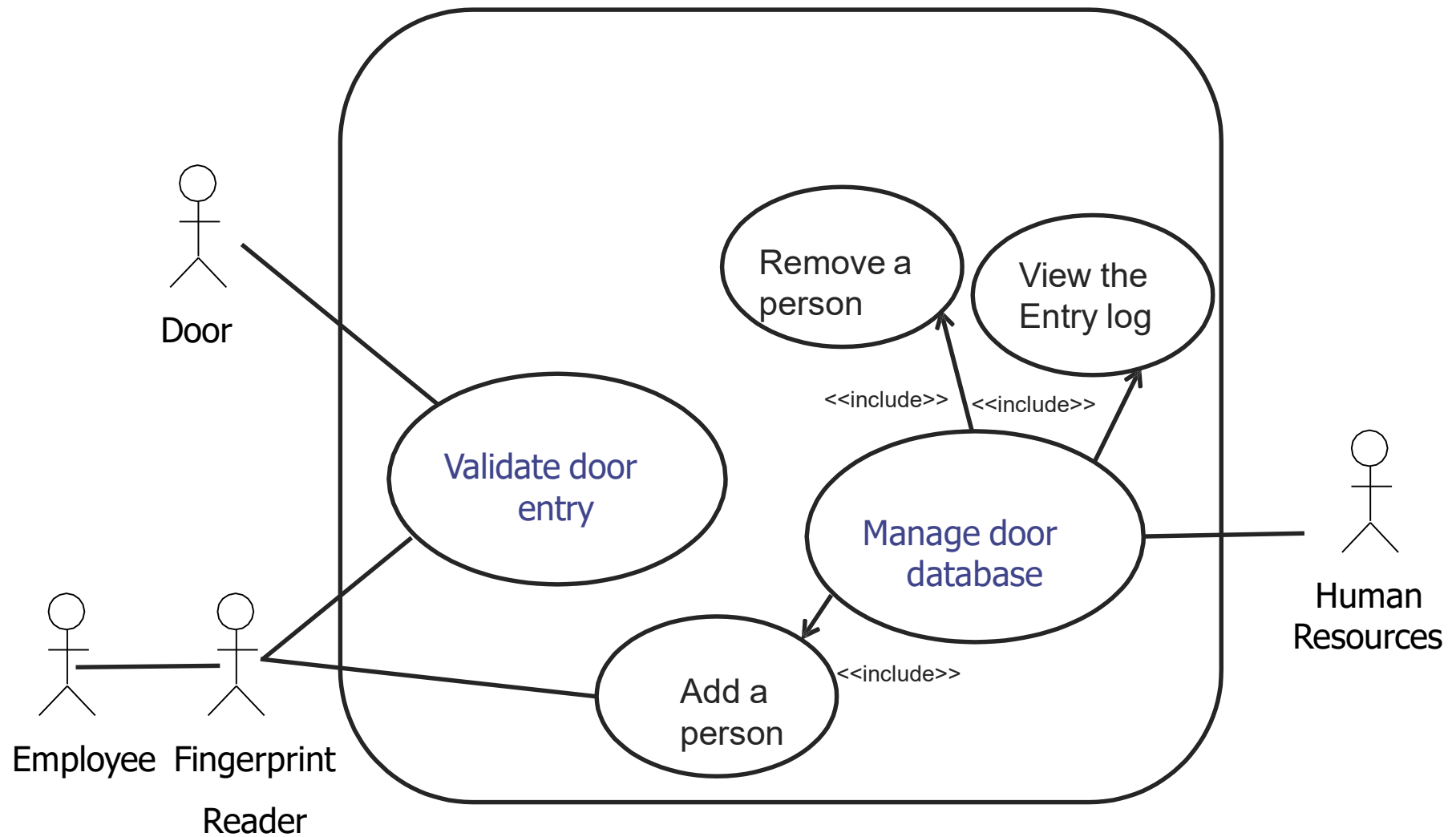
3 & 4. Refine Use Case diagram

REFINE THE MAIN USE CASES

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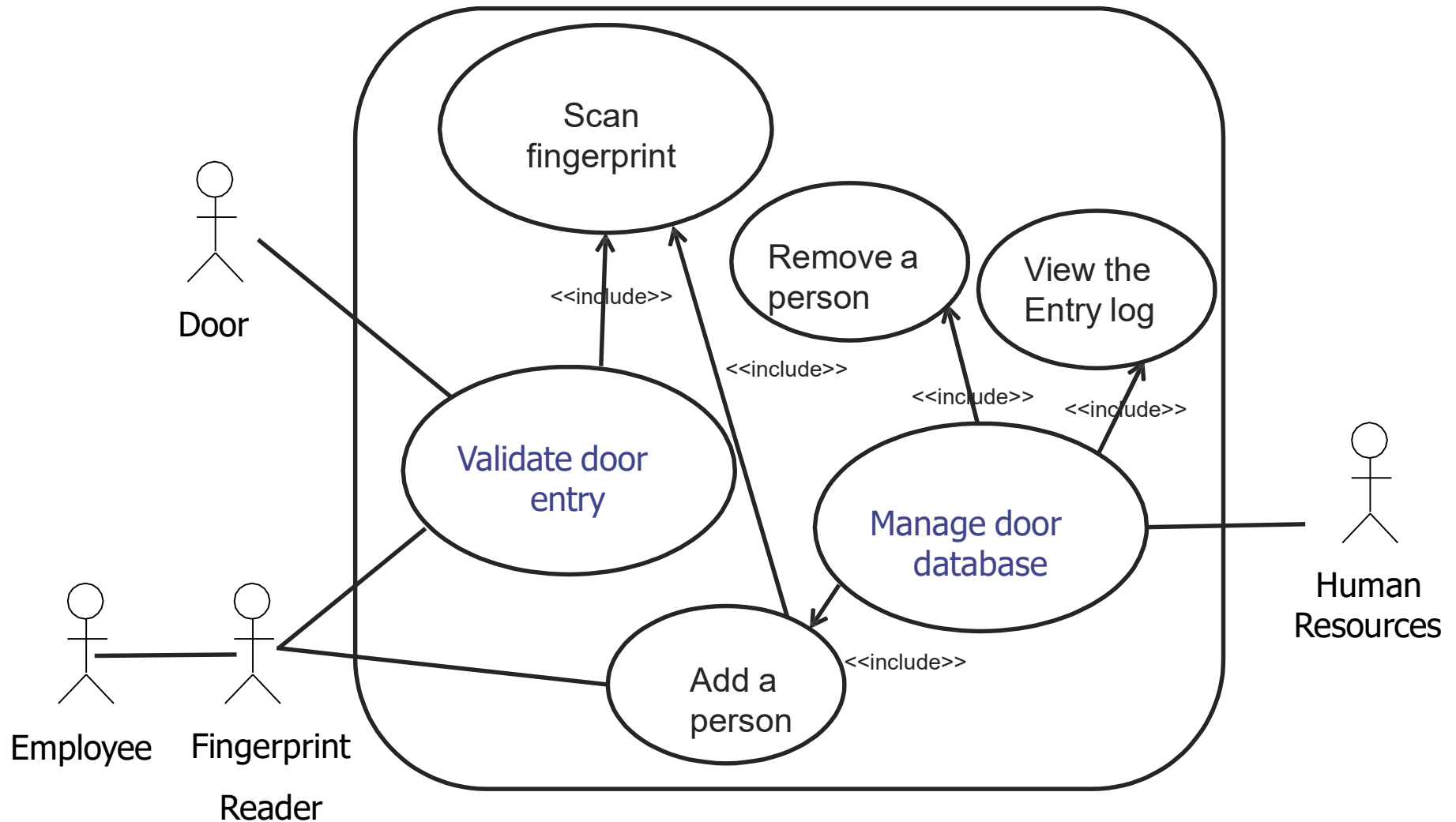
3 & 4. Refine Use Case diagram

REFINING THE MAIN USE CASES



3 & 4. Refine Use Case diagram

REFINING THE MAIN USE CASES



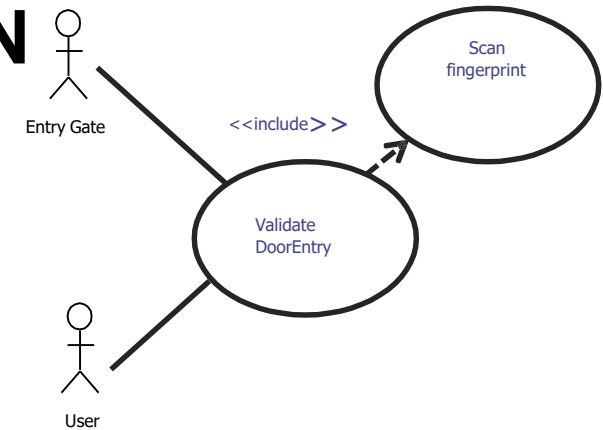
3 & 4. Refine Use Case descriptions

See the Use Case Description Template available on NTULearn course website.

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Frequency of Use:			
Flow of Events:			
Alternative Flows:			
Exceptions:			
Includes:			
Special Requirements:			
Assumptions:			
Notes and Issues:			

EXAMPLE USE CASE DESCRIPTION

ValidateDoorEntry Use Case Description



Flow of Events:

1. Employee presses finger on FingerprintReader.
2. FingerprintReader scans fingerprint (using use case Scan fingerprint) and sends image to the FacilityAccessManager.
3. The FacilityAccessManager validates the fingerprint
4. If fingerprint is valid display the message "Please enter" and unlock the EntryGate.
 - 4.1 The EntryGate sends a signal to the FacilityAccessManager that it has opened and closed.
 - 4.2 The FacilityAccessManager locks the EntryGate, and logs the entry time and name of the employee who entered in the entry log.

Alternative Flow:

- AF-S4. If fingerprint is not valid display the message "Access denied" and log the attempted entry time in the entry log.

Example: A security door access system

QUESTIONS THAT ARISE AND NEED CLARIFICATION WITH THE CLIENT

1. What is the detailed process when a fingerprint is validated?

What if the gate is opened and no one enters?

What if the gate is opened and does not close?

Is there a timeout?

What if more than one person enters?

2. Is there a particular format the Employee data is to be recorded in the log?

Does it need to be exported to another system or printed?

Other questions will arise as detailed Use Cases are written. These need to be clarified with the customer reflected in changes in the Use Case diagrams and descriptions

Summary:

Requirement Elicitation and Analysis Process

